EX.NO : 04

DATE : 22/07/2026

STUDY OF LAN CONFIGURATION

AIM :-

To design, set up, & configure a Local Area Network (LAN) connecting four computers through a network switch, assign static IP addresses to all devices.

EQUIPMENTS REQUIRED :-

S.No	Equipment	Quantity	Specifications.
1)	Personal computers (PC)	4	Ethernet (NIC) enabled.
2)	Network switch	1	8/16/24 port, unmanaged / managed
3)	Ethernet cables (Cat5e/Cat6)	4	Straight - through, RJ-45 connectors.

IMPLEMENTATION IN CISCO PACKET TRACKER :-

Step-1 : Place the Devices

1) open cisco packet Tracer

2) From the device category panel, select End devices & drag 4 PCs onto the workspace.

3) Select switches & drag 1 switch onto the workspace.

Step-2 : Cable the Devices

4) Click Connections → Select Copper straight-Through cable.

5) Click each PC, then click the switch, & select an available FastEthernet port to complete each connection.

6) Repeat for all four PCs (ports Fa0/1 to Fa0/4): Green link lights confirm a succesful physical connection.

Step-3 : Assign IP Addresses to PCs

7) Click a PC → Desktop tab → IP configuration

8) Enter the static IP Address & subnet Mask as per the IP table for each PC.

Step-4 : Configure the switch Management IP

9) Click the switch → CLI tab.

10) Enter the following commands to assign a management IP :

```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

Step-5 : Verify connectivity using ping

11) Click a PC → Desktop tab → command prompt.

12) Use the ping command to test connectivity to the other PCs, e.g., ping 10.1.1.2.

13) Successful "Reply from ...." messages confirm connectivity.

Step-6: Demonstrated Data Sharing

14) Add a server device & connect it to the switch.

15) Assign it an IP address via its Desktop → IP configuration tab.

16) Open the server's services tab & enable the HTTP service.

17) From another PC, open the Web Browser and enter the server's IP address to load the page, confirming successful data transfer across the LAN.

Step-7: Observe Packet Flow in Simulation Mode

18) Switch to simulation mode.

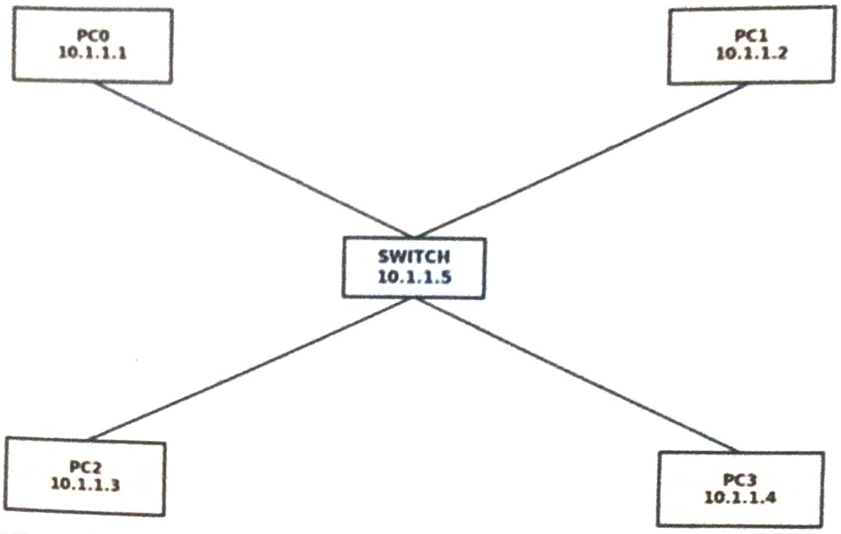
19) Send a ping or web request b/w two devices & use the play/capture controls to step through the packet as it travels hop-by-hop through the switch, observing MAC-address-based forwarding.

IP ADDRESS CONFIGURATION TABLE :-

Device	IP Address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch port FA0/1
PC1	10.1.1.2	255.0.0.0	switch port Fa0/2
PC2	10.1.1.3	255.0.0.0	switch port Fa0/3
PC3	10.1.1.4	255.0.0.0	switch port Fa0/4
Switch (Management)	10.1.1.5	255.0.0.0	VLAN 1 Interface

NETWORK DIAGRAM :-

Star Topology LAN (4 PCs + Switch)



RESULT :-

Thus the study & setup & configuration of LAN using a switch & ethernet cabel.

8/ 22/7/26